Mimosa-AI: evolving scientific workflows with Policy-BT

Code is the genotype. Workspace artifacts are the evidence. Selection and feedback drive the next generation.

Task goal + input data — Scientific tools via MCP · optional literature grounding

1 Synthesize a workflow

LLM → executable Python

Agents, prompts, tools and control flow
Seed / template during initialization

A → B → C → D

2 Execute the workflow

Sandboxed workflow runner

Reset workspace to the initial inputs
Run agents and scientific tools
Save scripts, logs, tables and figures

3 Policy-BT verifier

Preserve + reproduce

Owned snapshots; required .sh contract
Fresh-venv rerun enabled by default

Code-first pairwise comparison

Goal → fixed criteria + schema adapters
One cached deterministic Python policy
Relevant images may resolve code ties

Bradley–Terry tournament

All generation pairs in the task session
Refit every generation's current reward

Numeric rewards + evidence-grounded diagnoses

Reproduction failure → published reward 0

ratings

4 QD archive + parent selection

Refresh quality from current BT ratings
Quality + code novelty − length penalty
Retain survivors; select parent workflow(s)

parents

5 Mutation or crossover

Mutation: parent diagnosis → LLM directive
Crossover: recombine selected parent code
Use the selected parents' own feedback

Next-generation prompt

Code embedding → novelty

Selected-parent textual feedback → variation

Repeat until the configured stop condition or generation limit

Policy-BT auto mode: champion plateau, after the minimum number of generations.

stop

Restore the best workspace

Highest current BT rating—not the latest run

QD = quality–diversity. Numeric rewards guide selection; textual diagnoses guide changes. Benchmark grading is separate.